

DEXTER FERNANDES

Artificial Intelligence Engineer Bristol, UK +44 7436 007369 dexterfernandes11@gmail.com linkedin.com/in/dexterferns
github.com/Dexter-Fernandes

SUMMARY

Computer vision engineer with 3+ years shipping production ML systems in Python and PyTorch, owning model work end to end from data curation and training through deployment and field validation. Built object detection, segmentation and image-quality pipelines under real-time constraints, backed by REST APIs and containerised inference services linking models to live systems. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Built image-quality checks with OpenCV and scikit-image to detect mud and sand contamination on camera feeds, cutting false positives by 13%.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72h to under 24h.
- Designed and wrote a watchdog service's REST API, exposing service health and restart control so operators and other services could act on failures remotely.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.

LivSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review, foreground separation and failure analysis on industrial camera feeds.
- Automated image labelling and tagging by repurposing existing detection models, adding dataset versioning with DVC and cutting manual labelling effort by 80% on large real-world datasets.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames, cutting duplicate incident alerts from moving-vehicle feeds.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Designed a geometry-first cross-camera (MTMC) identity layer over per-camera DeepStream trackers, lifting pooled IDF1 from 0.198 to 0.245 on WildTrack.
- Built a CPU-testable MTMC association core using constrained union-find clustering and mutual-nearest-neighbour matching, reaching 100% online/offline agreement across 10,608 rows.
- Evaluated a DeepStream ReID SGIE using appearance embeddings for cross-camera identity matching against a geometry-only baseline, choosing geometry as the default after a 28% throughput cost returned only 0.002 F1 gain.
- Built an automated model-promotion gate checking match-rate and mean-IoU against a signed SHA-256 manifest, and wired GPU smoke plus MOT17 integration tests into GitHub Actions CI.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLOv26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.

- Containerised the stack with Docker on Linux and managed versioning in Git for reproducible builds across development environments.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-robot operation.
- Configured Nav2 in simulation for autonomous waypoint traversal over the generated map, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++, SQL, Bash
Computer Vision & DL	Object detection, image classification, semantic and instance segmentation, embedding-based similarity and re-identification, object tracking, data augmentation, CNNs, YOLO (v5/v8/v10/v26), PyTorch, TensorFlow, Keras, OpenCV, scikit-image, NumPy
Edge Inference & Optim.	TensorRT, OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, latency and throughput profiling
Video & Image Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-camera synchronisation, duplicate and identity-mismatch resolution, image-quality and artefact detection
Production & APIs	REST API design, containerised microservices, Docker, inference services, batch processing, CI/CD, GitHub Actions, pytest
Data & MLOps	Apache Airflow, Apache Kafka, MLflow, DVC (Data Version Control), CVAT, automated labelling and tagging, dataset versioning, synthetic data generation
Cloud & Databases	Microsoft Azure (Container Apps, Blob Storage, queues), PostgreSQL, autoscaling, cost-per-inference
Experimentation	Optuna, Weights & Biases, ablation studies, benchmarking harnesses, failure-mode analysis
Professional	Technical leadership, Agile and Scrum, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell — Service Champion / Team Trainer

09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)